

Sarvaraju Karthik

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Professional Summary

Software Engineer II with 4+ years of experience building production-critical banking and deposit-processing systems. Designed fraud/risk controls spanning **Self-on-Self validation** and **Risk Validation Adapter integration** with TLS/mTLS, standardized risk signaling, and transaction re-evaluation and restored customer-facing teller workflows in production. Strong in Java/Spring Boot, REST APIs, secure integrations, platform modernization, and production reliability.

Technical Skills

Languages: Java, Go, Python, JavaScript, TypeScript

Backend: Spring Boot, REST APIs, Microservices, Distributed Systems, API Design, Event-Driven Architecture, SQS, SNS, Concurrency, Caching

Databases: DynamoDB, MySQL, PostgreSQL, Oracle, Microsoft SQL Server (MSSQL), MongoDB, CognitoDB, Redis, SQL, Schema Design, Database Migrations, Indexing, Query Tuning

Cloud / DevOps: AWS (EC2, S3, Lambda, IAM, SES, ECS, OpenSearch/Elasticsearch), Google Cloud Platform, Google Secret Manager, Docker, Kubernetes, Istio, Jetty, Ehcache, Jenkins, CI/CD

Security / Observability: TLS/mTLS, Application Security, CVE Remediation, Secret & Certificate Management, Mend, JFrog Xray, SonarQube, Coverity, CloudWatch Metrics, CloudWatch Logs, Dashboards, Distributed Tracing, Production Debugging, RCA

Practices: System Design, High-Level Design (HLD), Low-Level Design (LLD), Multi-tenant Systems, Operational Excellence, Scalability Engineering, Agile, SDLC, Code Reviews, Documentation, Secure Coding

AI / Automation: Multi-Agent Systems, LangGraph, LangChain, RAG Pipelines, MCP Servers, LLM Integration, Cursor Skills

Experience

CANDESCENT

Software Engineer II

Hyderabad, India

Sept 2024 – Present

- Designed and delivered configurable **Self-on-Self deposit fraud validation** for CSPIImaging 4.2.0.x, owning **HLD/LLD**, account/routing comparison logic, configuration strategy, and ATM/Completion integration to support operator correction and re-validation while preserving backward compatibility.
- Strengthened deposit-risk decisioning by integrating the **Risk Validation Adapter (RVA)** with production-grade **TLS/mTLS**, fixing item-update re-evaluation defects, and exposing standardized **Red/Yellow/Green risk indicators** through platform APIs for downstream banking systems.
- Unblocked AppSec certification for PI 25.3 and PI 26.1** by remediating Mend/Xray/Coverity/Sonar findings, upgrading third-party/vendor dependencies, and porting security fixes across **3 repositories and 5+ supported release lines**.
- Remediated **100+ security vulnerabilities**, including **40+ Critical** and **60+ High-severity findings**, across multiple production services using **Mend, JFrog Xray, SonarQube, and Coverity**, resolving vulnerable dependencies and code-level security issues to improve release readiness and application security posture.
- Restored CSPI production Teller UI ID Scan stability** by preventing concurrent scan requests and correcting scanner cold-state/UI failures, eliminating application lockups affecting branch identity-verification workflows.
- Modernized platform runtime and cloud readiness through **Jetty 12.0.30, GCP Secret Manager/Istio**, Spring extended support, and **Ehcache 2.x to 3.x** migrations, keeping legacy and current banking release lines secure, supported, and cloud-deployable.
- Built end-to-end **decline-reason capture** across Review UI, REST APIs, and platform integration APIs, ensuring every rejected deposit carries structured reason data for **audit trails and operational reporting** on the 4.2.0 release line.
- Built custom **Cursor Skills** integrating OpenAI, Gemini, and Claude APIs to correlate application logs with recurring failure patterns and generate root-cause/remediation guidance, reducing developer log-triage effort from **45 to 15 minutes**.

NCR VOYIX (Formerly known as NCR CORPORATION)

Software Engineer I

Hyderabad, India

Jul 2022 – Aug 2024

- Contributed to **LLD and backend implementation of real-time ATM/ITM fraud validation**, integrating transaction processing, Rules, and Risk Validation services through REST APIs with configurable thresholds, persistent risk decisions, role-based review/override, and database support across **MSSQL, Oracle, and PostgreSQL**.
- Drove **JDK 8/11 to 17 migration across 8 services**, including a legacy monolith with **3K+ build issues**; resolved compatibility and resource-leak problems and completed production rollout with **zero downtime**.
- Stabilized the **CSPIImaging 4.0.1 hotfix line** by porting MICR/API fixes, resolving Java/JUnit build failures, adding missing database indexes, and remediating vulnerable dependencies, restoring the build pipeline for supported bank deployments.

Software Engineer Intern

Jan 2022 – Jul 2022

- Built a **UI automation suite from scratch** using JavaScript, designing modular and reusable functional components for repeatable testing while gaining hands-on experience with **Java, Spring Boot, REST services**, and Agile software delivery.

Projects

FlavorGraph — Graph-Based Recipe Discovery Platform

Live | GitHub

Java 21, Spring Boot, React, TypeScript, Neo4j/CognoDB, Cypher

- Built a full-stack recipe discovery platform using **Java 21, Spring Boot, React, TypeScript**, and **Neo4j/CognoDB**, implementing graph-based recipe ranking and multi-hop ingredient substitution through **REST APIs**.

Education

Neil Gogte Institute of Technology

Bachelor of Engineering in Computer Science and Engineering

Hyderabad, India

Aug 2018 – May 2022